DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM

Objectives

1. Apply COUNT(), SUM(), AVG(), MIN(), and MAX()
2. Generate total sales reports
3. Find top customers based on purchase amount
4. Identify best-selling products
5. Perform category-wise sales analysis

1. Aggregate Functions

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result



1.1 COUNT()

```
SQL> SELECT COUNT(Order_ID) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
           3
```

1.2 SUM()

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;

TOTAL_SALES
-----------
       7200
```

1.3 AVG()

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM Orders;

AVERAGE_ORDER_AMOUNT
--------------------
               2400
```

1.4 MIN()

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order
  2  FROM Orders;

MINIMUM_ORDER
-------------
         1500
```

1.5 MAX()

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order
  2  FROM Orders;

MAXIMUM_ORDER
-------------
         3200
```

2. Generate Total Sales Reports

```
SQL> SELECT
  2      COUNT(Order_ID) AS Total_Orders,
  3      SUM(Total_Amount) AS Total_Sales,
  4      AVG(Total_Amount) AS Average_Order_Value,
  5      MIN(Total_Amount) AS Minimum_Order,
  6      MAX(Total_Amount) AS Maximum_Order
  7  FROM Orders;

TOTAL_ORDERS TOTAL_SALES AVERAGE_ORDER_VALUE MINIMUM_ORDER MAXIMUM_ORDER
------------ ----------- ------------------- ------------- -------------
           3        7200                2400          1500          3200
```

2.1 Customer-wise Sales Report

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY
  9      c.Customer_ID,
 10      c.First_Name,
 11      c.Last_Name
 12  ORDER BY Total_Purchase DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
         103
Anitha R
          3200

         101
Harini N
          2500

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------

         102
Priya S
          1500
```

Purpose

This report shows:

• Customer ID

• Customer name

• Total purchase amount

• Customers arranged according to purchase amount

3. Find Top Customers Based on Purchase Amount

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY
  9      c.Customer_ID,
 10      c.First_Name,
 11      c.Last_Name
 12  ORDER BY Total_Purchase DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
TOTAL_PURCHASE
--------------
        103
Anitha R
          3200

        101
Harini N
          2500

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
TOTAL_PURCHASE
--------------

        102
Priya S
          1500
```

4. Identify Best-Selling Products

```
SQL> SELECT
  2       p.Product_ID,
  3       p.Product_Name,
  4       SUM(od.Quantity) AS Total_Quantity_Sold
  5  FROM Product p
  6  JOIN Order_Details od
  7       ON p.Product_ID = od.Product_ID
  8  GROUP BY
  9       p.Product_ID,
 10       p.Product_Name
 11  ORDER BY Total_Quantity_Sold DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       101
Face Wash
                   3

       103
Shampoo
                   2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

       104
Lipstick
                   2

       102
Moisturizer
```

Purpose

This query:

1. Connects Product and Order_Details.

2. Groups records according to product.

3. Calculates total quantity sold.

4. Displays products from highest to lowest quantity sold.

4.1 Product Sales Amount

```
SQL> SELECT
  2       p.Product_ID,
  3       p.Product_Name,
  4       SUM(od.Quantity) AS Total_Quantity_Sold,
  5       SUM(od.Subtotal) AS Total_Sales
  6  FROM Product p
  7  JOIN Order_Details od
  8       ON p.Product_ID = od.Product_ID
  9  GROUP BY
 10       p.Product_ID,
 11       p.Product_Name
 12  ORDER BY Total_Sales DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       104
Lipstick
                  2        3200

       101
Face Wash
                  3        3000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------

       103
Shampoo
                  2        1800

       102
Moisturizer
```

5. Category-wise Sales Analysis

```
SQL> SELECT
  2      c.Category_ID,
  3      c.Category_Name,
  4      SUM(od.Quantity) AS Total_Quantity_Sold,
  5      SUM(od.Subtotal) AS Total_Sales
  6  FROM Category c
  7  JOIN Product p
  8      ON c.Category_ID = p.Category_ID
  9  JOIN Order_Details od
 10      ON p.Product_ID = od.Product_ID
 11  GROUP BY
 12      c.Category_ID,
 13      c.Category_Name
 14  ORDER BY Total_Sales DESC;

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
          4
Makeup
                    2        3200

          1
Face Care
                    3        3000

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------

          3
Hair Care
                    2        1800
```

6. Combined Business Sales Report

```
SQL> SELECT
  2       c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3       o.Order_ID,
  4       p.Product_Name,
  5       cat.Category_Name,
  6       od.Quantity,
  7       od.Subtotal
  8  FROM Customer c
  9  JOIN Orders o
 10       ON c.Customer_ID = o.Customer_ID
 11  JOIN Order_Details od
 12       ON o.Order_ID = od.Order_ID
 13  JOIN Product p
 14       ON od.Product_ID = p.Product_ID
 15  JOIN Category cat
 16       ON p.Category_ID = cat.Category_ID
 17  ORDER BY o.Order_ID;

CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
Harini N
       201
Face Wash

CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------
```

Report Structure

This report can be used to understand:

- Who purchased the product
- Which order was placed
- Which product was purchased
- Which category the product belongs to
- Quantity purchased
- Amount generated